Lecture plan · for the lecturer

Evaluating agents

AI agents · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Define pass@k.
- Quantify uncertainty.
- Grade evaluation rigour.

Activity: Audit an evaluation

1. Warm-up (5 min): Show two result tables and ask students which they believe.
2. Compute (15 min): Pairs compute pass@1 and pass@3 from a sample of runs.
3. Interval (15 min): Add 95% intervals and decide which differences are real.
4. Audit (20 min): List what each team's report omits and rewrite one honestly.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- How wide was the interval?
- Which omission mattered most?
- How will you weight rigour?

Exit ticket

- Write the one sentence every agent result should include.

Try it in your next lecture
Show two teams' results, one inflated and one honest, without saying which; pairs list what each did and did not report, then vote on which they trust.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **Which result is most trustworthy for a student agent project?**

◯ pass@1 over 5 runs on held-out tasks ◯ pass@10 on development tasks ◯ Best of one run

2 **Compute the metrics.**

With p = 0.5, pass@3 = _____ . On 100 tasks at $\hat{p}$ = 0.5, the standard error is _____ .

Word bank: 0.75 · 0.05 · 0.875 · 0.5

3 **Which make an agent evaluation more trustworthy? Choose all that apply.**

☐ Reporting only the best run ☐ Reading transcripts of successes

☐ Several runs with an interval ☐ Tasks held out from development

4 **If agent A scores 62% and agent B 58% on 50 tasks in one run, the gap may be only noise.**

◯ True ◯ False

5 **Match each threat to evaluation validity with its remedy.**

Tuning on the test tasks	• • • •	Several runs and intervals
Run-to-run randomness	• • • •	Fresh, unpublished tasks
Agent edits the tests	• • • •	A held-out set
Benchmark in training data	• • • •	Protected checkers and transcript review

Answers · for the lecturer

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1 **pass@1 over 5 runs on held-out tasks**

Held-out, repeated and honest.

2 **With p = 0.5, pass@3 = 0.875. On 100 tasks at $\hat{p}$ = 0.5, the standard error is 0.05.**

All three tries fail with probability 0.5^3, so pass@3 = $1 - 0.5^3$ = 0.875.

3 **Several runs with an interval; Tasks held out from development; Reading transcripts of successes**

Best-of reporting inflates results.

4 **True**

The interval is about ±14 points; a 4-point gap is within noise.

5 **Tuning on the test tasks — A held-out set**
Run-to-run randomness — Several runs and intervals
Agent edits the tests — Protected checkers and transcript review
Benchmark in training data — Fresh, unpublished tasks

A held-out set stops tuning on the test, several runs with intervals measure randomness, protected checkers and transcript review catch an agent that edits tests, and fresh unpublished tasks avoid contamination.

For you to think about

How do your students currently report results from agent or model projects, and what would you require next term?